LQR Control of Vehicle Bicycle Model

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I. Introduction

Path following is a critical task which autonomous vehicles must complete in order to safely navigate in their environment. Autonomous driving architecture typically consists of a perception, planning, and control pipeline[1]. The perception stack uses sensors to detect objectives or obstacles in the world, the planning stack receives data from perception and uses that data to calculate a safe path for the vehicle to traverse, and the control stack takes the path and sensor data and calculates control signals used to drive the vehicle on the path.

Currently, controls for path following is mostly done with conventional control methods including Proportional Integral Derivative (PID) [2], Linear Quadratic Regulator (LQR)[3] or Model Predictive Control (MPC)[4]. Reinforcement learning algorithms, including Deep Deterministic Policy Gradient, are also being researched for control in path following applications [5].

The objective of this work is to implement a Linear Quadratic Regulator controller for path following and velocity tracking.

A. Notation

x_w, y_w	World coordinate system
$X,\ Y$	Vehicle center of mass position (global coordinate frame)
$x_t, y_t / x_h, y_h$	Trailer axle / Hitch point position
$\dot{x},\ \dot{y}$	Velocity components (in vehicle coordinate frame)
$\alpha_f,\ \alpha_r$	Front and rear tire slip angles
v_f, v_r	Front and rear axle velocity vectors
v	Velocity of vehicle center of mass
δ	Steering angle
$l_f,\ l_r,\ l_t$	Front and rear distance from axle to center of mass, trailer length
ψ	Vehicle yaw angle (in global coordinate frame)
$\dot{\psi}$	Vehicle yaw rate
$F_{xr}, F_{yr}, F_{xf}, F_{yf}$	Forces in the x/y direction on the front/rear tires
I_z	Vehicle yaw moment of inertia
$C_{\alpha f},\ C_{\alpha r}$	Front and rear tire cornering stiffness
$C_d,\ \rho,\ A$	Drag coefficient, Air density, Cross section area

II. Modeling

It is important to use an accurate model of the system when designing a controller so that if the controller is deployed into a physical system, the physical response will be similar. The model of a vehicle that will be used is a linearized bicycle model [6] with a simple Pacejka tire model [7].

A. Vehicle Bicycle and Pacejka Magic Tire Model

The Vehicle Bicycle Model and Pacejka Magic Tire Model were chosen as a simplified model for the complex vehicle dynamics. Additionally, drag force was added to the model to provide a realistic representation of longitudinal resistance, ensuring the vehicle's behavior is more aligned with real-world dynamics and to prevent unbounded acceleration during simulation scenarios. The diagram of the Vehicle Bicycle Model is shown below in Figure 1.

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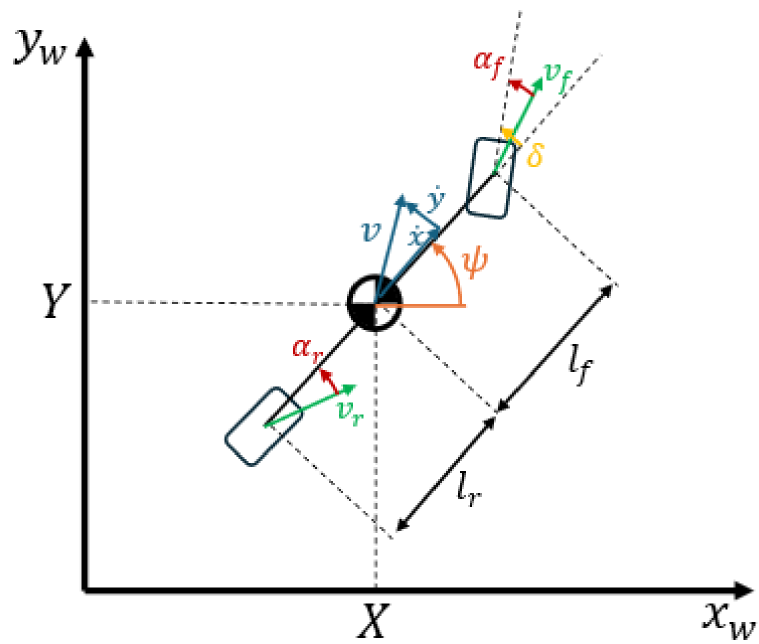

Fig. 1: *Vehicle Model Diagram*

The cornering stiffness of the front and rear tires, $C_{\alpha f}$ and $C_{\alpha r}$, are used for a simple linear approximation of the Pacejka tire model. For small slip angles $\alpha, F_{yf} \approx C_{\alpha f}\ \alpha_f$ and $F_{yr} \approx C_{\alpha r}\ \alpha_r$. The slip angles and corresponding tire forces are calculated below.

$$\alpha_f \approx \delta - \frac{\dot{y} + l_f\dot{\psi}}{\dot{x}} \ \rightarrow\ F_{yf} \approx C_{\alpha f}(\delta - \frac{\dot{y} + l_f\dot{\psi}}{\dot{x}})$$

$$\alpha_r \approx -\frac{\dot{y} - l_r\dot{\psi}}{\dot{x}} \ \rightarrow\ F_{yr} \approx C_{\alpha r}(-\frac{\dot{y} - l_r\dot{\psi}}{\dot{x}})$$

Using the above equations for F_{yf} and F_{yr}, setting $F_x = F_{xr} + F_{xf}$, and using $F_{drag} = \frac{C_d\ \rho\ A\ \dot{x}^2}{2}$ we obtain the following.

$$\ddot{x} = -\frac{C_d\rho A\dot{x}}{2m}\dot{x} + \frac{C_{\alpha f}\sin\delta}{m\ \dot{x}}\dot{y} + \frac{C_{\alpha f}\sin\delta l_f}{m\ \dot{x}}\dot{\psi} + \frac{1+\cos\delta}{m}F_x - \frac{C_{\alpha f}\sin\delta}{m}\delta$$

$$\ddot{y} = -\frac{C_{\alpha r} + C_{\alpha f}\cos\delta}{m\ \dot{x}}\dot{y} + \frac{C_{\alpha r}\ l_r - C_{\alpha f} l_f\cos\delta}{m\ \dot{x}}\dot{\psi} + \frac{\sin\delta}{2\ m}F_x - \frac{C_{\alpha f}\cos\delta}{m}\delta$$

$$\ddot{\psi} = \frac{l_r C_{\alpha r} - l_f\ C_{\alpha f} cos\delta}{I_z\dot{x}}\dot{y} - \frac{l_f^2 C_{\alpha f} cos\delta + l_r^2 C_{\alpha r}}{I_z\dot{x}}\dot{\psi} + \frac{l_f sin\delta}{2\ I_z}F_x + \frac{l_f\ C_{\alpha f} cos\delta}{I_z}\delta$$

$$x_h = X - l_r cos(\psi)$$

$$y_h = Y - lrsin(\psi)$$

$$\dot{\psi}_t = \frac{\dot{x}}{l_t}\sin(\psi_t - \psi)$$

$$\psi_t \mathrel{+}= \dot{\psi}_t \Delta T$$

$$x_t = x_h - l_t cos(\psi_t)$$

$$y_t = y_h - l_t sin(\psi_t)$$

The inputs to the system are the applied force (F_x) on the vehicle, and the steering angle (δ). In practice, neither of these inputs can be directly set; the applied force would be dependent on the tire/road friction and the output torque from the motors, which would be controlled by the input voltage and current. The steering angle cannot be instantaneously set, and instead the rate of change of the steering angle would be controlled. The outputs of the model are the velocity in the x and y directions ($\dot{x},\ \dot{y}$), and the yaw rate ($\dot{\psi}$). These are all in the vehicle reference frame and can be transformed into the global reference frame to estimate the position ($X,\ Y$) and heading (ψ) of the vehicle.

B. Kinematic Trailer Model

Let the tractor have pose (x_t, y_t, ψ_t) and longitudinal velocity v_t. The hitch point, located a distance l_r behind the tractor reference point, is given by

$$x_h = x_t - l_r \cos\psi_t, \tag{1}$$
$$y_h = y_t - l_r \sin\psi_t. \tag{2}$$

The trailer axle is positioned a distance L behind the hitch, with pose (x, y, ψ). Define the articulation angle

$$\alpha = \psi_t - \psi. \tag{3}$$

Under the pure rolling (no lateral slip) assumption, the kinematic equations of motion for the trailer are

$$\dot\psi = \frac{v_t}{L}\sin(\psi_t - \psi), \tag{4}$$
$$\dot x = \dot x_h - L\dot\psi \sin\psi, \tag{5}$$
$$\dot y = \dot y_h + L\dot\psi \cos\psi, \tag{6}$$

where the hitch velocities are obtained from the tractor motion:

$$\dot x_h = v_t \cos\psi_t, \tag{7}$$
$$\dot y_h = v_t \sin\psi_t. \tag{8}$$

Substituting these into the trailer equations yields

$$\dot x = v_t \cos\psi_t + L\dot\psi \sin\psi, \tag{9}$$
$$\dot y = v_t \sin\psi_t - L\dot\psi \cos\psi, \tag{10}$$

with $\dot\psi$ given by (4).

C. Linearized State Space Model

The system presented above is non-linear and must be linearized. The system will be linearized about $\delta = 0$, and $\dot x = v$ where v is a constant reference velocity.

$$\begin{bmatrix}\ddot x\\ \ddot y\\ \ddot\psi\end{bmatrix} = \begin{bmatrix}\frac{-C_d\ \rho\ A\ v}{2m} & 0 & 0\\ 0 & \frac{-(C_{\alpha r}+C_{\alpha f})}{m\ v} & \frac{l_r C_{\alpha r} - l_f C_{\alpha f}}{m\ v}\\ 0 & \frac{l_r C_{\alpha r} - l_f C_{\alpha f}}{I_z\ v} & \frac{-(l_r^2 C_{\alpha r} + l_f^2 C_{\alpha f})}{I_z\ v}\end{bmatrix}\begin{bmatrix}\dot x\\ \dot y\\ \dot\psi\end{bmatrix} + \begin{bmatrix}\frac{1}{m} & 0\\ 0 & \frac{C_{\alpha f}}{m}\\ 0 & \frac{l_f\ C_{\alpha f}}{I_z}\end{bmatrix}\begin{bmatrix}F_x\\ \delta\end{bmatrix}$$

D. Discrete Time State Space Model

The symbolic representation of the discrete time system is too large to be practical to use. Instead, the vehicle parameters are substituted into the continuous time system above. The vehicle parameters used are taken from the Tesla Model S [8] and can be seen in Table 1. A reference velocity of $v = 15$ was used as well.

m	1500 kg	**Lf**	1.2 m
Iz	3000 kg m^2	**Lr**	1.6 m
Cf	80000 N/rad	**Cd**	0.208
Cr	80000 N/rad	**A**	2.4 m^2
Ts	0.1 s	ρ	1.225 kg/m^3

TABLE I: *Parameters table*

Once all of the parameters were substituted in, the zero order hold discritization method was used to obtain the following discrete time system.

$$\begin{bmatrix}\dot x\\ \dot y\\ \dot\psi\end{bmatrix}_{k+1} = \begin{bmatrix}0.999 & 0 & 0\\ 0 & 0.494 & 0.0699\\ 0 & 0.0349 & 0.494\end{bmatrix}\begin{bmatrix}\dot x\\ \dot y\\ \dot\psi\end{bmatrix}_k + \begin{bmatrix}6.66 & 0\\ 0 & -3.678\\ 0 & 2.173\end{bmatrix}\begin{bmatrix}F_x\\ \delta\end{bmatrix}_k$$

E. System Properties

Important properties about the system are given here:
Transfer Matrix

$$G(z) = \begin{bmatrix} \frac{6.66\cdot 10^{-5}z^2 - 6.58\cdot 10^{-5}z + 1.60\cdot 10^{-5}}{1.0z^3 - 1.98z^2 + 1.22z - 0.241} & 0 \\ 0 & \frac{-3.67z^2 + 5.64z - 1.96}{1.0z^3 - 1.98z^2 + 1.22z - 0.241} \\ 0 & \frac{2.17z^2 - 3.37z + 1.20}{1.0z^3 - 1.98z^2 + 1.22z - 0.241} \end{bmatrix}$$

Smith McMillan Form:

$$\begin{bmatrix} \frac{6.66\cdot 10^{-5}z^2 - 6.58\cdot 10^{-5}z + 1.60\cdot 10^{-5}}{1.0z^3 - 1.98z^2 + 1.22z - 0.24} & 0 \\ 0 & \frac{1.0}{1.0z^3 - 1.98z^2 + 1.22z - 0.24} \\ 0 & 0 \end{bmatrix}$$

Zero Polynomial:

$$6.66 \cdot 10^{-5}z^2 - 6.58 \cdot 10^{-5}z + 1.60 \cdot 10^{-5}$$

Pole Polynomial:

$$3.94 \left(0.50z^3 - z^2 + 0.61z - 0.12\right)^2$$

Poles:

$$\{0.44 : 2,\ 0.54 : 2,\ 0.99 : 2\}$$

Zeros:

$$\{0.44 : 1, 0.54 : 1\}$$

Smith-McMillan Degree: 6

III. Controllability

A. State Feedback Control

It was determined that the system is controllable, and therefore stabilizable, by testing the rank of the controllability matrix C_{ab}. The eigenvalues of the system are also already stable, and the states of the system would eventually decay to zero if no control input was given. Pole Placement was used to design a state feedback controller responsible for driving the states to zero faster though. The eigenvalues used are $\lambda = -1, -2, -3$, and the resulting F is shown here.

$$F = \begin{bmatrix} 45002.291 & 14462.404 & 24306.808 \\ -0.000208 & -83.945 & -139.774 \end{bmatrix}$$

The stabilization of the model was tested by simulating the non-linear model driven by the control law $u = -Fx$. The simulation results can be seen in Figure 2 below.

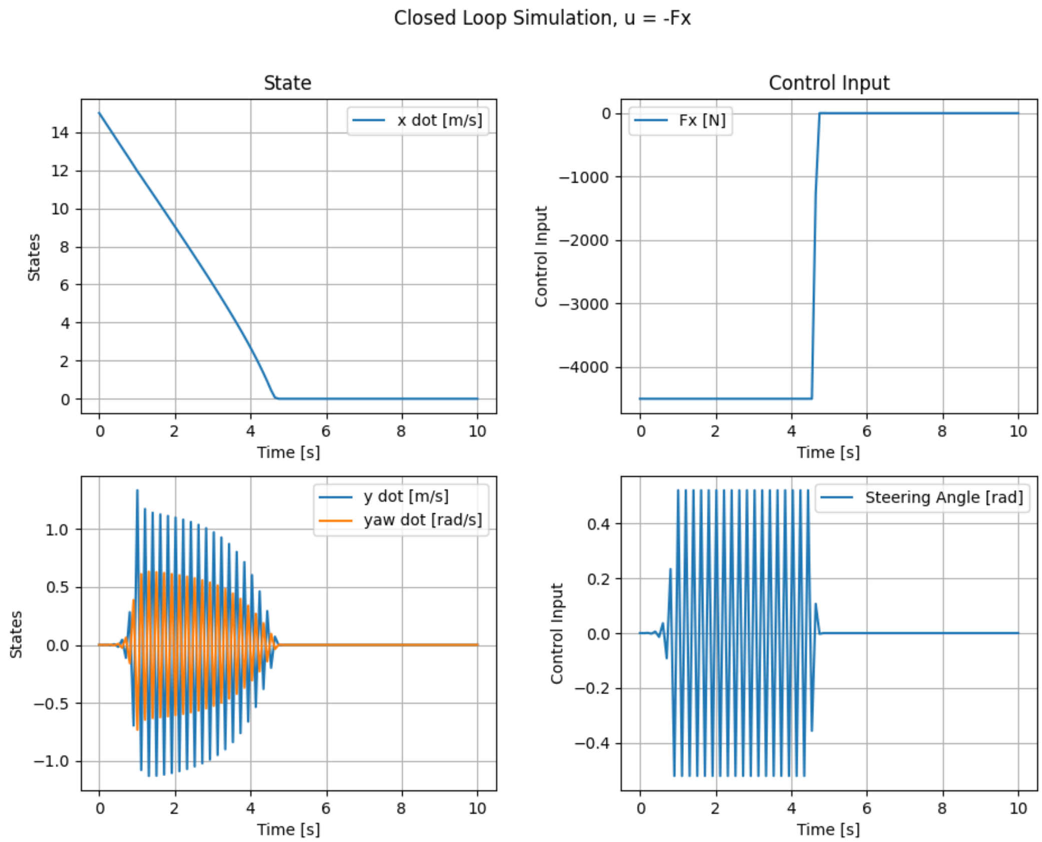

Fig. 2: *Closed Loop Simulation for State Feedback Control*

All states are eventually driven to zero, however, this method uses extremely high control signals in order to achieve this. A physical vehicle would be bounded by maximum signals it can act on. The simulation above takes this into account, but the model is still frequently using the maximum allowed control signal.

Another problem with this controller is the coupling between control signals. Although all three states are coupled in the nonlinear vehicle model, speed and steering are mostly independent of each other, which can be seen in the linearized model. A better controller would be in form presented below, which decouples the steering and speed control signals.

$$F = \begin{bmatrix} f_{11} & 0 & 0 \\ 0 & f_{22} & f_{23} \end{bmatrix}$$

B. Jordan Form

The Jordan Form of the A matrix is:

$$A_{JF} = \begin{bmatrix} 0.999 & 0 & 0 \\ 0 & 0.444 & 0 \\ 0 & 0 & 0.543 \end{bmatrix}$$

IV. Observability

The model uses the velocities of the vehicle as the state variables. For this system, the positions of the vehicle can not be recovered using a linear observer, and the model is technically unobservable. However, it is possible to track the global positions of the vehicle using discrete time integration, as long as the initial conditions are known. The equations used to track the global position of the vehicle are:

$$X \;=\; X \;+\; \dot{x}\; cos(\psi)\Delta t - ysin(\psi)\Delta t$$

$$Y \;=\; Y \;+\; \dot{y}\; cos(\psi)\Delta t + xsin(\psi)\Delta t$$

$$\psi \;=\; \psi \;+\; \dot{\psi}\; \Delta t$$

V. Output Regulation

A. Problem Setup

The output regulation problem was initially set up so that the controller implemented would drive the nonlinear vehicle model along a sinusoidal trajectory. The desired trajectory was generated using the exosystem and can be seen below.

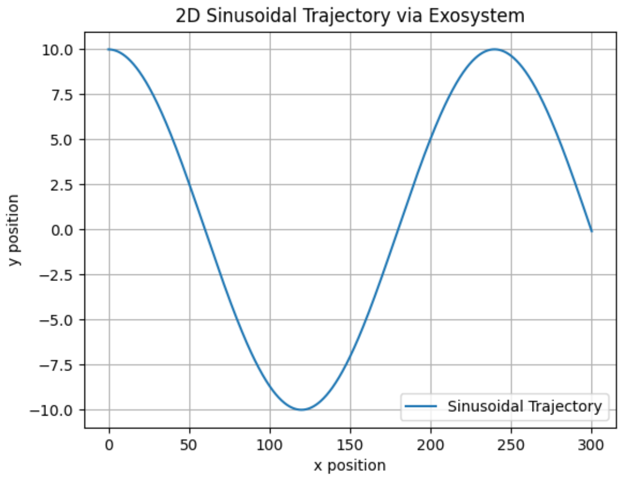

Fig. 3: *Sinusoidal Trajectory Generated with Exosystem*

The matrix used to generate this trajectory is A_2 with initial conditions $x_2(0)$, The remaining matrices required for output regulation design are also presented below. Q and R were experimentally determined, it was found that a very large value was needed for q_{11} for the controller to have a chance at tracking the desired velocity. Furthermore, r_{22} was chosen to be 10 to penalize steering control and resulted in smoother trajectories taken by the vehicle.

$$A_2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\omega^2 & 0 \end{bmatrix}, x_2(0) = \begin{bmatrix} 0 \\ \dot{x}(0) \\ amplitude \\ 0 \end{bmatrix}$$

$$C_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, D_1 = 0$$

$$Q = \begin{bmatrix} 100000 & & & & & & \\ & 2.5 & & & & & \\ & & 10 & & & & \\ & & & 0 & & & \\ & & & & 0 & & \\ & & & & & 0 & \\ & & & & & & 0 \end{bmatrix}, R = \begin{bmatrix} 0.001 & 0 \\ 0 & 10 \end{bmatrix}$$

The final controller gain is presented below. Since this problem is formatted for trajectory tracking, the error (reference - state) was passed to the controller to calculate the control signal instead of passing the state directly. The simulation showing the controller driving the non-linear vehicle model are shown below.

$$K = \begin{bmatrix} 13867 & 0 & 0 & 0.480 & 0.249 & 0 & 0 \\ 0 & -0.945 & 2.209 & 0 & 0 & 0.394 & 0.0185 \end{bmatrix}$$

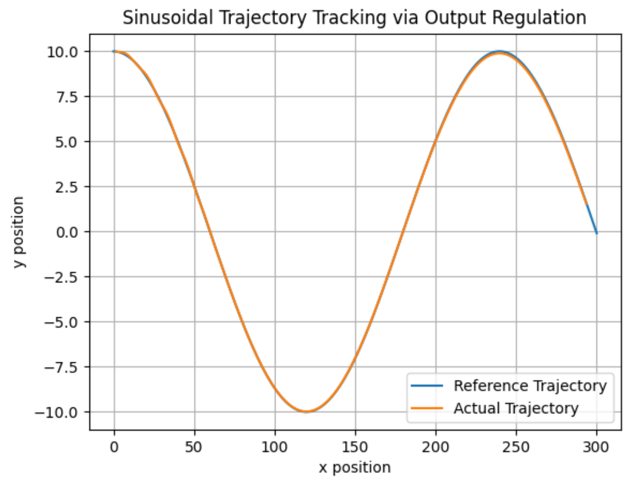

Fig. 4: *Vehicle Trajectory Compared to Reference Trajectory*

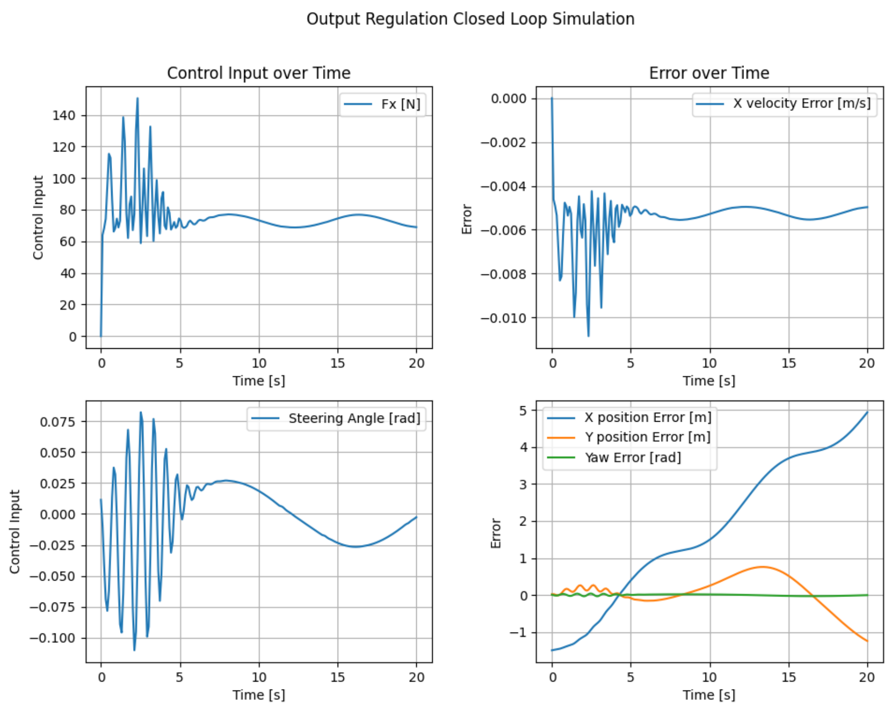

Fig. 5: *Errors and Control Signals Recorded in Output Regulation Controller Simulation*

The controller developed was able to achieve very low levels of yaw error, but the error in the longitudinal velocity, and the x and y positions was suboptimal. There was clear steady state error in the longitudinal velocity, likely caused by the drag force on the vehicle. This steady state error caused the x position error to continuously increase. It was not immediately clear if there was steady state error for y positional error, but it was clear that there was room for improvement in trajectory tracking.

The controller also resulted in high frequency oscillating control signals at the beginning of the simulation, which would be difficult to implement in a real vehicle and likely unsafe for passengers.

VI. Integral Derivative Control with LQR

To test the controllers ability to reject disturbances and generalize to different paths, the controller designed above using the sinusoidal exosystem was deployed into a new environment where the path was generated using a cubic spline that connected three randomly generated points. The resulting path was similar in nature to the sinusoidal path, but made the problem more realistic because the true path was not known ahead of time.

Another important change with the new environment is replacing the trajectory x and y error with cross-track error, the distance between the vehicle's center of gravity and the closest point on the path. This slight distinction changes the problem from a trajectory tracking problem to a problem combining path following and velocity tracking. A properly tuned controller will perform the same with either lateral error definition, but switching to cross-track error introduces more decoupling between the control signals and makes it easier to manually tune controller parameters.

Although the controller designed to track sinusoidal paths performed well when tracking the original path, it did not work well when deployed into the new environment with a random path. The controller over steered resulting in oscillations around the path, and was not able to track the reference speed at all. An example of the controller driving the non-linear vehicle in the random environment is shown below. Note that the y-axis scaling is different from the figures presented previously. The y-axis in all of the images below goes from -100 meters to +100 meters. The x-axis remains the same, from 0 meters to 300 meters.

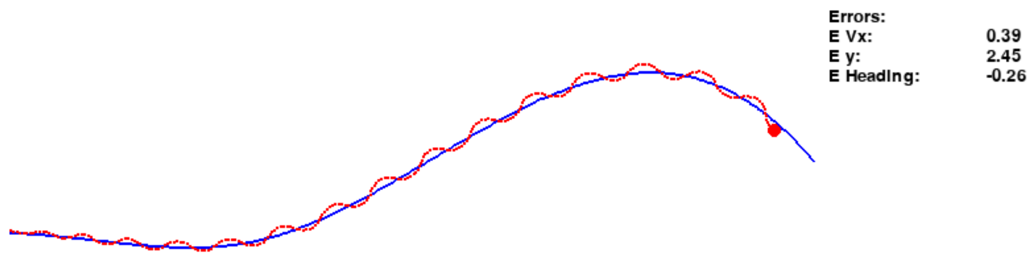

Fig. 6*: Performance of Original Controller Designed with Sinusoidal Exosystem Output Regulation*

In addition to the controller designed in the Output Regulation section, a simple LQR controller was designed using the A and B matrices from the linearized model. The Q and R matrices used were the same as in output regulation, with the intention of biasing the controller towards larger longitudinal force control and smaller and smoother steering control. The K matrix for the LQR controller was found using the MATLAB command lqr(A, B, Q, R). A sample simulation of the simple LQR controller driving the vehicle is shown below.

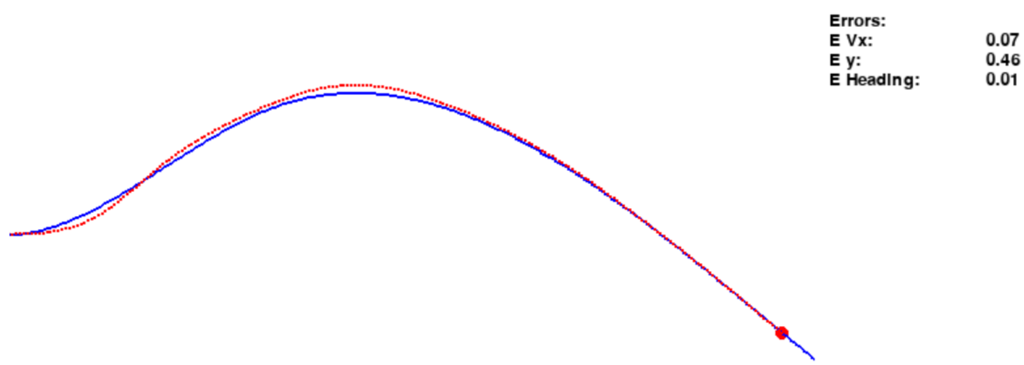

Fig. 7*: Sample Simulation of Simple LQR Controller*

The simple LQR controller performed better than the Output Regulation controller, but had steady-state error for speed control and path tracking. This controller also did not perform well in turns where the resulting cross-track error and speed error were significantly higher. To correct this, integral and derivative control was added for all errors. The block diagram for the new system architecture is shown below.

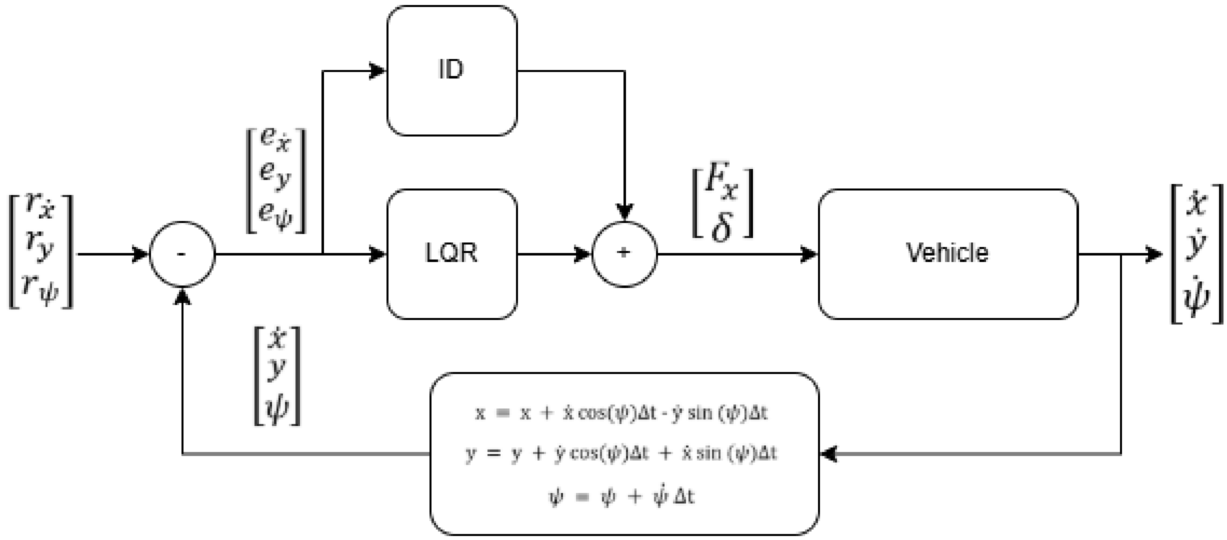

Fig. 8: *Control Diagram Adding Integral and Derivative Control*

Integral control corrected the steady-state error in the longitudinal velocity by offsetting the effects of the drag force. However, the integral control on the y position error and yaw error introduced oscillations around the path. Derivative control was then added to dampen the oscillations.

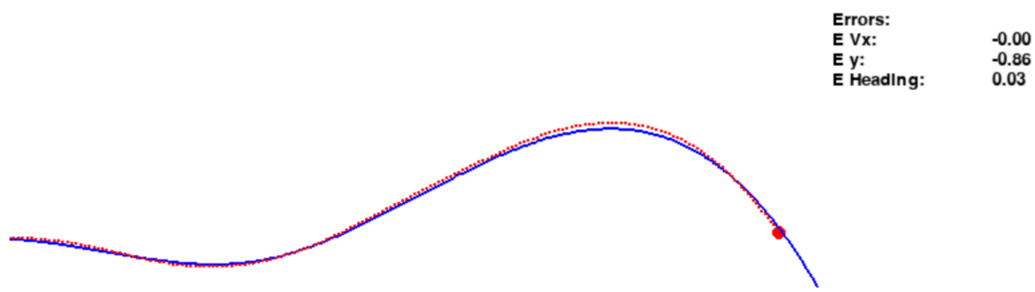

Fig. 9: *LQR Controller with Integral and Derivative Control*

The same approach of introducing integral and derivative control was applied to the controller developed for output regulation. A high derivative gain of 10 was applied to both steering states, which helped reduce the magnitude of oscillations. A sample simulation with this controller is shown below.

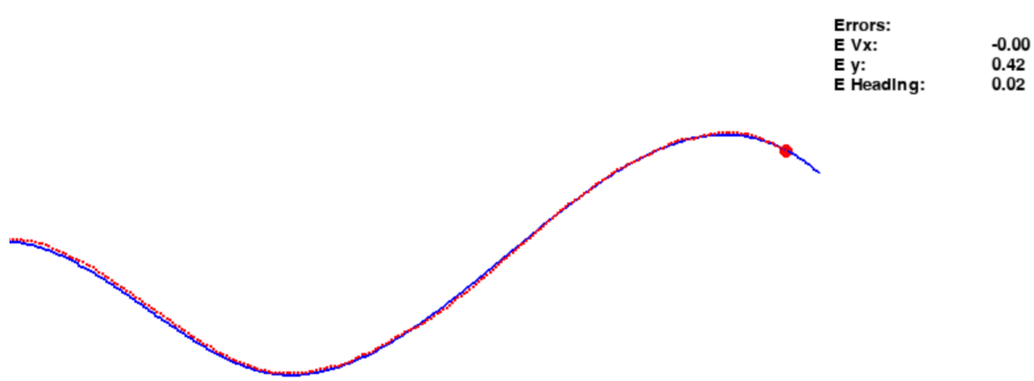

Fig. 10: *Output Regulation with Sinusoidal Exosystem and Integral and Derivative Control*

VII. Conclusion

At first glance, the final controller presented combining the output regulation controller with integral and derivative control seems to have the best performance. The steady state error is lower than all of the other controllers and it tracks the path very well, but it still has problems with steering where the steering angle oscillates at a very high frequency. This high frequency oscillation is not possible to implement on a real vehicle, because steering angle can not be set directly, and the rate of change of the steering angle should be controlled instead. Even if it was possible, it would not be safe or comfortable to implement on a real vehicle.

The controller with the best performance, that is feasible to implement in a real vehicle is the LQR controller with integral and derivative control. Although it does not track the path perfectly, it steers smoothly and stays reasonably close to the desired path.

Unfortunately, none of the controllers designed are accurate enough to be safely driven on a Canadian highway. The Geometric Design Guide for Canadian Roads suggests a minimum lane width of 3.7 meters [9]. The Tesla Model S is 1.9 meters wide[8], leaving 1.8 meters of space in the lanes. If the desired path was in the middle of the lane, then a maximum of 0.9 meters error is acceptable. All of the controllers displayed error higher than 0.9 meters at times, especially in sharp turns. It is important to note that the paths generated included much sharper turns than what would be experienced in a highway. All of the controllers performed significantly better on straighter paths, and most of the error was observed in sharp turns.

References

[1] "Perception, planning and control for self-driving system based on on-board sensors," *ResearchGate*, October 2024. 1

[2] Z. Wang *et al.*, "Driverless simulation of path tracking based on pid control," *IOP Conference Series: Materials Science and Engineering*, vol. 892, no. 1, p. 012050, July 2020. 1

[3] Z. Wang, K. Sun, S. Ma, L. Sun, W. Gao, and Z. Dong, "Improved linear quadratic regulator lateral path tracking approach based on a real-time updated algorithm with fuzzy control and cosine similarity for autonomous vehicles," *Electronics*, vol. 11, no. 22, January 2022. 1

[4] P. Stano *et al.*, "Model predictive path tracking control for automated road vehicles: A review," *Annual Reviews in Control*, vol. 55, pp. 194–236, January 2023. 1

[5] G. Hess and W. Ljungbergh, "Deep deterministic path following," April 2021, arXiv:2104.06014. 1

[6] B. Varma, N. Swamy, and S. Mukherjee, "Trajectory tracking of autonomous vehicles using different control techniques (pid vs lqr vs mpc)," in *2020 International Conference on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE)*, October 2020, pp. 84–89. 1

[7] H. B. Pacejka, *Tire and Vehicle Dynamics.* Oxford: Butterworth-Heinemann, 2012. 1

[8] "Tesla model s owner's manual," 2024, [Online]. Available: https://www.tesla.com/ownersmanual/models/en_us/GUID-91E5877F-3CD2-4B3B-B2B8-B5DB4A6C0A05.html. 3, 10

[9] T. A. of Canada (TAC), *Geometric Design Guide for Canadian Roads*, Ottawa, 2020. 10